

Assignment 1

Scientific Machine Learning: Blending Science with Data

Question 1 (5 marks)

Consider the two equations below

$$(x + 1)^2 = -(y - 5)^2 + 8 \quad (1)$$

$$(y - 4)^2 = -x^3 \quad (2)$$

- a) Write a Julia program to plot the two equations above.
- b) By visual inspection of your plot in part a), specify how many pair(s) (there could be just 1 pair or more than 1 pair) of (x, y) values that simultaneously satisfy both equations. Write down approximately what they are.
- c) Write a Julia program that uses the Newton-Raphson method to solve the two equations above. Use your guess in part b) as your initial starting point(s). To get full marks for this question, you need to find all the solutions. You should use the Julia `NonlinearSolve` package to check your answer

Question 2 (7 marks)

Consider the function given below

$$f(x_1, x_2) = -x_1 x_2 e^{-(x_1^2 + x_2^2)/2} \quad (3)$$

- a) Plot this function using the contour function in Julia
- b) Just from your plot, and without performing any computation, write down approximately where you think the local minima (there are more than 1) are.
- c) Write a Julia program using standard gradient descent technique to find the local minima of the function
- d) Run your program start with the initial guess $x_0 = (-2, 2)$, can you find a local minima using this initial guess?

- e) Run your program start with the initial guess $x_0 = (-2, -2)$, can you find a local minima using this initial guess?
- f) Explain your results in d) and e). You are encourage to provide relevant graphs to accompany your explanation.

Question 3 (8 marks)

You are given temperature data (taken via thermocouples) from a 10m rod. The rod has temperature 40°C at $x = 0$ and the temperature is approximately 10°C at $x = 10\text{m}$. The rod is subjected to 3 different heating conditions. You are asked to find a model for the temperature distribution on the rod.

Use the linear least squares technique or the gradient descent approach or any of the other methods taught in lectures (see Jupyter Notebooks 3.1-3.5) to find the best models for the data in the file 'Q3Data.csv'.

There are 3 different data sets in this file corresponding to the 3 different heating conditions. The data is stored in a CSV file with 6 columns with pairs of data under the following headers "xQ1a", "yQ1a", "xQ1b", "yQ1b", "xQ1c", "yQ1c". Each two columns are a dataset.

For every one of the three dataset, you will need to

- a) Plot the data and see how it looks like
- b) Decide what the best functional form for the mathematical model
- c) Decide on which (of the two, least squares or gradient descent) method to use to find the coefficients so that your model best fit the data.

To read the data, you can use the following code segment

```
df = CSV.read("Q3Data.csv", DataFrame;
select=["xQ1a", "yQ1a",
"xQ1b", "yQ1b",
"xQ1c", "yQ1c"])
x_samples=df[:, :xQ1a]
y_samples=df[:, :yQ1a]
```

You will need to install the CSV and DataFrames packages to use the code above.